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Detection of DDoS Attack Using Machine Learning Techniques

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***Abstract*— Distributed Denial-of-Service (DDoS) attacks are one of the worst network security threats facing the world today. DDoS attacks disable online services by bringing servers under attack with large volume of malicious traffic, making the services inaccessible to legitimate users. Traditional detection approaches are lacking due to the changing techniques being used by attackers. In an effort to provide a solution for this problem, machine learning (ML) methodologies have been identified as good candidates due to their ability to process large volume of traffic data and identify anomalies in real-time. In presented paper, various machine learning algorithms including Decision Trees, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Neural Networks. are applied to protect against DDoS attacks, The system utilized supervised learning to classify network traffic as normal or malicious. The experimental findings reveal that Neural Networks and SVM yield high accuracy and efficient detection rates in comparison with other algorithms, whereas Decision Trees are faster but less accurate. The results show that machine learning-based solutions are effective in enhancing the detection and mitigation of DDoS attacks and provide a reliable and scalable solution for network system protection.**

Keywords*— *Network security, Supervised learning, Machine learning, DDoS attack, SVM, Neural Networks, Detection, and Decision Trees.

I. INTRODUCTION

In digital era, cyberattacks are increasing in frequency and threatening businesses, governments, and individuals. DDoS attack is a major concern as it A DDoS attack functions by overwhelming a targeted server, or service with huge amount of suspicious traffic, making the service inaccessible to legitimate users. The impacts like interruption to service, money loss, consumer confidence, and reputational losses are common sign of DDoS attacks.

High-profile targeted attacks in notable organizations within the recent past have created immense requirements for advanced mechanisms of detection as more effective as well as being more efficient in nature. Traditional detection methods, like signature and rule-based cannot handle the changing tactics used by attackers.

To address these challenges, researchers and cyber security practitioners are working on machine learning (ML) techniques as a potential alternative. Machine learning relies on data-driven models to uncover anomalies and classify malicious behaviour in network traffic. Unlike traditional measures, ML techniques have the capability to learn from new data repeatedly, enabling them to keep pace with new evolving patterns of attacks. Since machine learning can detect subtle and advanced anomaly patterns, it acts as an effective countermeasure for DDoS detection. Some ML algorithms including Decision Trees, Support Vector Machines (SVM), K-Nearest Neighbor (KNN), and Neural Networks are evaluated and applied to detect DDoS attack. All have some pros and cons and it is important to select the best model based on accuracy, computational cost, and scalability. Decision Trees are intuitive and simple, while SVM provides strong classification in high-dimensional spaces. KNN excels in pattern recognition, while Neural Networks are designed to deal with big and complicated data.

Although machine learning deals with massive data, reducing false alarms, and receiving detection in real time. Data preprocessing plays an important role to improve data quality and performance. By comparing the advantages and limitations of each algorithm, we proposed a reliable detection system that can aid in protecting the network from DDoS attacks.

A. Research Contribution

Proposed study presents a robust and efficient ML-based model to detect DDoS attack. Parameters are tuned to generate more efficient and accurate results. With advanced preprocessing methods the performance of model is improved. Through the various testing and comparative analysis, shows the remarkable performance of model better than the traditional methods for correct classification. This work enhances the state of DDoS detection and provides valuable insights for the development of reliable, real-world network security solutions.

II. LITERATURE REVIEW

DDoS attacks present the most critical security challenges to network, affecting businesses and individuals. The attacks manifest themselves in an overwhelmingly large amount of malicious traffic that makes services or systems subject to the attack inaccessible to bona fide users. With

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advancements in DDoS attacks, the conventional techniques like signature-based and rule-based schemes are ineffectual.

Peneti et al. [1]. introduced a machine learning-based DDoS attack detection method focusing on feature selection and classification to enhance detection performance. Their work showed that preprocessing and feature engineering are important aspects of improving model performance Nalayini et al. [2] investigated several ML algorithms, including Decision Trees, Random Forest, and SVM, to detect various patterns of DDoS attacks with less false positives.

Author [3] proposed a model for DDoS detection using machine learning. Another study [4] analyzed the network security from DDoS utilizing ML based algorithms. Author [5] investigated various ML models to detect DDoS in software defined network. The light weighted ML models can deliver a balanced trade-off between accuracy and computational overhead. Author [6] proposed a study for DDoS using machine learning and fuzzy logic.

Author [7] proposed a study for DDoS using studied ML-based techniques achieving major performance factors like detection accuracy and flexibility. Zekri et al. [8] investigated ML-based detection methods tailored for cloud computing environments, facing scalability issues and introducing an integrated ML framework for real-time detection of DDoS. Sarraf, Saman. [9] investigated different ML classifiers and emphasized the significance of feature selection and optimization methods in enhancing DDoS detection accuracy. Their research focused on the effectiveness of ensemble learning models.

Saini et al. [10] suggested a combination of ML classifiers to boost resistance against advanced variations of DDoS attacks, introducing ensemble techniques to improve detection accuracy. Wani et al. [11] explored ML-based detection approaches specifically designed for cloud computing environments and their effectiveness in counteracting attacks and response time reduction. Ashfaq et al. [12] presented a real-time zero-trust model for the detection of DDoS attacks in 5G edge networks based on ML in SDN environments.

III. METHODOLOGY

The proposed method contains various phases to detect DDoS attacks, which starts from data collection to preprocessing, feature extraction, model selection, training and testing the performance. All steps contribute to creating a good and reliable detection system. The machine learning based approach is applied for systematic operation and reliable performance. The detailed of steps are mentioned below:

A. Data Collection

DDoS attack network logs dataset [13] is utilized in proposed study as shown in Figure1, contains real network traffic logs with normal and DDoS attack patterns. It is best suited for training machine learning models due to its diversity and uniqueness.

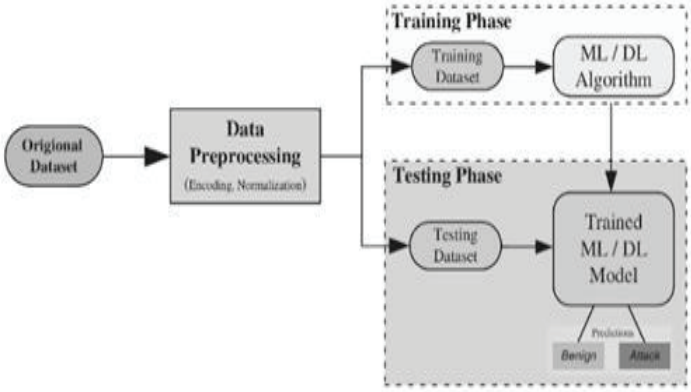

Fig. 1. Proposed Methodology

B. Data Preprocessing:

The raw data supplied in the dataset is normally noisy and may have missing values and inconsistencies. Table 1 shows the features used in proposed study. We processed and cleaned the data by doing the following:

- **Data Cleaning:** Eliminated any duplicated or incomplete records for ensuring data consistency.
- **Normalization:** Normalized the feature values between the range 0 and 1 using Min-Max scaling.
- **Encoding Categorical Features:** Transformed categorical data, such as protocol types, to numerical representation utilizing one-hot encoding.
- **Data Balancing:** Addressed the issue of class imbalance using the Synthetic Minority Over-sampling technique (SMOTE) and create a balanced dataset.

TABLE I. FEATURES USED

Feature Name	Description
SRC_ADD	Address of source
DES_ADD	Address of destination
PKT_TYPE	Type of packet
PKT_SIZE	Packet size
SEQ_NUMBER	Sequence number of packet.
NUMBER_OF_PKT	Total number of packets in the transmission.
PKT_RATE	Rate at which packets are sent.
PKT_DELAY_NODE	Delay caused by the node in processing the packet.
PKT_SEND_TIME	Time when the packet was sent.
PKT_CLASS	Classification of the packet

C. Feature Selection:

To enhance the performance of model, feature importance analysis is performed. These techniques enable to identify the most critical features that played a major. By reducing the dimension of features, the model became more effective and time-efficient at training and prediction.

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D. Model Selection:

Different machine learning algorithms are applied to identify the most suitable one for identifying DDoS attacks. The following algorithms were tested:

- **Decision Trees:** Simple and fast but likely to overfit if not appropriately tuned.
- **Support Vector Machines (SVM):** Well-suited for high-dimensional data binary classification.
- **K-Nearest Neighbors (KNN):** Computationally expensive for large data but well-fitted for pattern recognition.
- **Neural Networks:** Are able to learn to detect intricate patterns relatively well but with a price in terms of increased computing resources.

E. Model Training and Testing:

Pre-processed data were divided into the train and test sets based on k-fold cross-validation for avoiding overfitting. Hyperparameter was adjusted for optimal performance by Grid Search and Random Search. Model performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC curve metrics.

F. Model Selectiont:

The best-performing model, chosen on basis of accuracy and stability, was selected as a real-time DDoS detection system. The system continues to be in active surveillance of network traffic and sends notifications whenever suspicious traffic is detected, thereby enabling prompt action against imminent threats.

Utilizing the Kaggle dataset, an efficient model was developed to detect DDoS in real world.

Algorithm: 1

Step 1: Import Libraries

It starts by importing the necessary libraries such as numpy, pandas, and sklearn. These libraries assist in data handling, mathematical operations, and constructing the SVM model.

Step 2: Load the dataset.

Import your data into the system. The dataset usually consists of:

- **Features (X):** The attributes or features of the data.
- **Labels (Y):** The categories or results that we wish to predict.

Step 3: Data Preprocessing

1. **Data Cleaning:** Remove missing or invalid data points.
2. **Feature Scaling:** Standardize the data to have a consistent scale, like normalizing values between 0 and 1.
3. **Splitting the Data:** Divide the data into two parts- training and testing.

$$D = D_{train} + D_{test}$$

- **Training Set:** For training the model.
- **Testing Set:** For testing the performance of the model.

Step 4: Model Training

1. **Kernel Selection:** Select an appropriate kernel function depending on your data, e.g.:
   - **Linear Kernel:** Suitable when data is linearly separable.
   - **Polynomial Kernel:** For non-linear data with polynomial relationships.
   - **RBF (Radial Basis Function) Kernel:** Suitable for complex, non-linear data.
2. **Optimum Problem:** The primary task of SVM is to determine the optimum line (or hyperplane) that separates classes with the best margin.

$$min\frac{1}{2}||w||^2 + C\sum \xi i$$

   - **w:** Weight vector
   - **C:** Regularization parameter (tunes trade-off between margin size and misclassification errors)
   - $\boldsymbol{\xi i}$: Slack variables to accommodate data points that don't lie exactly on the line
3. **Decision Boundary:** The model determines the best line (or hyperplane) that separates the classes.

Step 5: Prediction

After training the model, use it to predict:

$$\hat{y} = \text{sign}(w \cdot x + b)$$

- **w:** Weight vector
- **x:** Input feature
- **b:** Bias term

Step 6: Model Evaluation

To get an idea of how well the model is performing, compute the following metrics:

1. **Accuracy:** Quantifies how frequently the model is accurate.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \times 100 \quad (1)$$

1. **Precision:** Focuses on how many of the predicted positives are actually positive.

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

2. **Recall:** Measures how many actual positives are correctly identified.

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

3. **F1 Score:** Balances precision and recall.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

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IV. RESULT & ANALYSIS

The performance of all the models was assessed and the models initially were not performing well were optimized and fine-tuned for better performance. The focus was to improve detection accuracy and reducing false positives to ensuring the development of robust models.

Machine learning model comparison was performed with a 70:30 train-test ratio, and the performance metrics were compared on Accuracy, Precision, Recall, and F1 Score. Accuracy

Among the five models, the accuracy of 96.02% was achieved by Support Vector Machine (SVM) model. Random Forest (RF) and KNN came next with accuracies of 96.18% and 96.51%, respectively. GBC and LR models were also very accurate with 95.75% and 94.12%, respectively. This is an indication that RF and KNN are very dependable in coming up with accurate predictions of the outcomes in the provided dataset.

A. Precision

The RF model achieves the precision 96.61% precision, and SVM with 95.93%. GBC and LR models also have high precision rates of 93.42% and 91.87%, respectively.

B. Recall

The recall figures demonstrate that KNN and GBC have a strong ability to correctly classify positive instances with figures of 88.72% and 89.56%, respectively. RF and SVM possess recall scores of 81.84% and 81.23%, respectively, while LR retains a recall of 88.67%. These findings indicate that KNN and GBC are better at classifying relevant instances than RF and SVM.

C. F1 Score

The F1 Score achieves a balance of precision and recall is obtained with the GBC model at 91.45%, then with LR at 90.12%. RF and KNN models yield equivalent F1 Scores at 85.67% and 89.28%, respectively. SVM yielded the lowest F1 Score of 84.89%. With such a high F1 Score for GBC, it must be achieving optimal precision and recall.

D. Overall Analysis

Random Forest is the most precise model, with Gradient Boosting Classifier having the best F1 Score, a measure of even performance between recall and precision. KNN possesses high accuracy with relatively lower precision compared to other models. Logistic Regression has a consistent competitive performance in all measurement metrics. Support Vector Machine model is precise with a relatively low F1 Score, which points to a sacrifice between precision and recall. Figure 2 shows the performance comparison of different models.

Based on the above observations, Random Forest and Gradient Boosting Classifier are most suitable for this problem, out of which RF is better at accuracy and GBC offers an equilibrium approach. The choice would depend on application needs—either accuracy or how much precision compared to recall must be achieved.

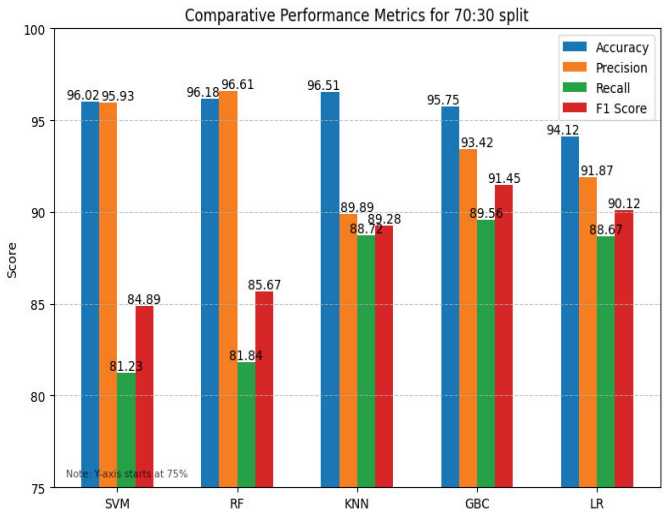

Fig. 2. Comparative Performance Metrics for SVM, Random Forest, and KNN in DDoS Attack Detection

V. CONCLUSION

Proposed work presents the DDoS attacks detection model using various machine learning algorithms. Through vast experimentation and analysis, we have concluded that RF and KNN proved to be better with 96.18% and 96.51% accuracy, respectively. These models had impressive performance regarding accuracy, recall, and F1 score, and were also adequate for real-world DDoS attack detection. While other models also generate competitive accuracy. RF provides an acceptable accuracy of 96.18%, was plagued by the problem of computational efficiency. GBC and LR were also providing acceptable accuracy (95.75% and 94.12%, respectively) but were slightly lower in recall and F1 measures than SVM and RF. These results further support the advantages of choosing appropriate algorithms in light of the particular requirement of precision, accuracy, and computational complexity.

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